



BAHIRDARUNIVERSITY
BAHIRDARINSTITUTE OF TECHNOLOGY
FACULTY OF MECHANICAL AND INDUSTRIAL ENGINEERING
INDUSTRIAL ENGINEERING PROGRAM

Course Guide Book: Strength of Materials I

1. GENERAL INFORMATION OF THE COURSE

Course Name:	Strength of Materials I
Course Code:	Meng2111
Course Credits Points:	5hrs.
Contact Hours:	2hrs lecture, 2hrs tutorial & 1hr laboratory
Course Module:	Mechanics of Materials
Course Category:	Major
Status of Course:	Compulsory
Pre-Requisites:	Engineering Mechanics I - (CEng1051)
Students Department:	Mechanical Engineering
Degree Program:	BSc. In Mechanical Engineering
Students Admission:	Regular
Year of Study:	2 nd
Semester:	First Semester
Academic Year:	2024/2025
Course Duration:	From March 2025 to June 2025
Course Chair:	Amare Mahtsentu
Course Instructor:	Dawit Molla dawitml101@gmail.com Office: Agri.orTana#217
Consultation:	Monday 10:00 AM -11:00 AM Friday 2:00 PM-3:30 PM

2. COURSE DESCRIPTION

Method of Sections, Stress, Axially Loaded Members, Shear Force and Shear Stress, Bearing Stress, Deformation and Strain, Stress-Strain Diagram, Hooke's Law and Deformation, Ultimate and Allowable Stress, Factor of Safety, Shearing Force and Bending Moment in Beams, Flexural Stress in Beams, Composite Beams, Shearing Stress in Beams, Deflection of Beams, Torsion, Struts and/or Column.

3. OBJECTIVES OF THE COURSE

- Analyze the stress-strain relationship in various structural members subjected to single loadings; axial load, torsion, bending and shear load.
- Solve the problems in various structural members subjected to combined loadings; a combination of axial load, torsion, bending and shear load.
- Design for dimension and strength of structural members subjected to various external loads, determine their deformation, and select the suitable material for a specific engineering application.

4. COURSE LEARNING OUTCOMES (CLO)

Attend of the course, students will be able to:

No	Course Learning Outcome (CLO)	Performance Indicators (PIs)	Program Student Outcome (PSO)
1	Analyze the stress-strain relationship in various structural members subjected to single loadings; axial load, torsion, bending and shear load.		3,1
2	Solve the problems in various structural members subjected to combined loadings; a combination of axial load, torsion, bending and shear load.		3,1,3
3	Design for dimension and strength of structural members subjected to various external loads, determine their deformation, and select the suitable material for a specific engineering application.		1,2

5. THE COURSE TIME SCHEDULE AND TEACHING METHODOLOGY

The following topics will be covered in this course with in this semester.

Table1: Time Schedule and Teaching Methodology of the course

Time Schedule	Course Contents	Teaching Methodology and References	Attaining student outcome
Week 1-2	1. Chapter One: Simple Stress and Strain Tension and Compression members; Stress and Deformation; Hook 's law; Stress-strain diagram;	• Seven Examples • Lectures supported by tutorials • 1 lab Demonstration	Outcome:1
Week 3-4	2. Chapter Two: Shearing Force and Bending Moment in Beams Application of method of chapters; Shear force and bending moment equations; Shear force and bending moment diagrams	• Four Examples • Lecture, Tutorial, discussions • 1 lab exercise • Textbook 1,2	Outcome:1,2
Week 5-6	3. Chapter Three: Bending Stress in Beams The Flexure Formula; Beams of two Materials.	▪ At least five examples ▪ Lecture, Tutorial, discussions, lab ▪ Ref.2, Textbook1and2	Outcome:3, 4
Week 7-8	4. Chapter Four: Slope and Deflection of Beams Strain-curvature and Moment Curvature Relations; Governing Differential Equations for Deflection, Solution by Direct Integration; Moment area Method	▪ At least three examples ▪ Lecture, Tutorial, discussions, lab ▪ Ref.2, Textbook 1and 2	Outcome:2, 4
Week9	Mid-Exam		
Week 10-11	5. Chapter Five: Shearing Stresses in Beams Relation Between Shear Force and Bending Moment	▪ At least 3 examples ▪ Lecture, Tutorial, discussions, lab ▪ Ref.2, Textbook1and2	Outcome:2, 4
Week 12-13	6. Chapter Six: Torsion of Circular Shafts Torsion Formula; Angle of Twist; Shearing Stresses and Deformation of Circular Shafts.	▪ Lecture, Tutorial, discussions, lab ▪ Ref.2, Textbook1and2	Outcome:2, 4
Week 14-15	7. Chapter 7: Complex Stresses: Compound stresses; Analysis of plane stress and strain: Principal stresses and maximum shear stress; Transformation of stress; Mohr 's circle of stresses and strains; Principal axes and principal moments of inertia.	Lectures supported by tutorials and Lab	

Week 16-17	8. Chapter 8: Struts and Columns Stability of equilibrium; The Euler formulas for different end restraints; Analysis of beam columns; The Secant formula; Eccentric loading.	Lectures supported by tutorials and Lab	
Week	Final Exam		

9. ASSESSMENT AND EVALUATION

The course utilizes a variety of continuous assessment and evaluation methods to ensure the successful achievement of course learning outcomes. These methods include assignments, projects, quizzes, midterm exams, final exams, homework, classwork, presentations, and more. It is imperative that all assignments and tasks are completed within the specified deadlines. If you encounter any difficulties with the assignments or tests, please contact the instructor promptly for assistance.

Home works, Assignments and project: Assignments may involve collaborative work for brainstorming and developing general approaches. However, unless specified otherwise, each individual must complete their own submissions. Assignments and projects may entail both individual and group components, with equal participation expected from all group members. While collaboration is encouraged, submitting another individual's work as one's own constitutes academic dishonesty and will incur in penalties. All group members must contribute equally. Timely submission is mandatory, as late submissions will not be graded. Oral exams will accompany assignments and projects, requiring thorough preparation from all members. Additionally, the instructor reserves the right to assign additional tasks or assignments as necessary.

Quizzes: Quizzes serve to assess students' current understanding of the coursework. They are conducted in class following lecture or tutorial sessions.

Mid-term Exam and Final Exam: The exams will be closed book. During exam all necessary materials will be provided by examiner.

Table2: Summary of Assessments

	Assessment type	Mark allotted	Content Coverage	Attaining course learning outcome (CLO)	Time Schedule
1	Assignments	15%	Chapter1,2,3,4,5	1,2,3,4	2
2	Laboratory Report	10%	Chapter 2,3,6,8	1	3
3	Mid Exam	25%	Chapter 1,2,3,4	1,2,3,4	--
5	Final Exam	50%	Chapter5,6,7,8,	1,2,3,4	16
	Total	100%			

10. LABORATORY/WORKSHOP/SESSION

Table3: Summary of laboratory/Workshop/Session content and required material

No.	Schedule	Laboratory Topic	Short Description of laboratory	Attaining course outcome
1	Week5	SFD & BMD	Shear force and bending moment diagram	1,2,3,4
2	Week8	Deflection	Deflection of beam	3,4
3	Week11	Torsion	Torsion of circular shaft	2,4
4	Week14	Column	Sturt buckling unit	2,4

11. COURSE POLICY

All students are expected to adhere to the code of conduct for students as per the Senate Legislation of Bahir Dar University, December 2020, during the duration of this course. Academic misconduct, which includes cheating, fabrication, and plagiarism, will not be tolerated and will be promptly reported to the relevant authorities for appropriate action.

The teaching methodology for this course will be diverse, encompassing activities such as reading, lectures, tutorial, discussions, homework, classwork exercises, and presentations, etc. Active participation from students is expected, including asking questions and raising issues whenever clarification is needed.

Regular attendance and punctuality are expected from all students. Those who miss more than 15% of the semester's classes will not be eligible to sit for the final exam. Attendance will be recorded for each class, and students are responsible for all announcements and changes made in class. Any student who exceeds the 15% absence threshold will receive a grade of 'NG' (No Grade) and must provide acceptable reasons for their absence, as per Senate legislation, Art. 174 for NG management.

Cell phones brought to class must be set to silent mode to avoid disruption to the teaching-learning process. Any form of disturbance during class time is unacceptable and must be avoided.

All students are required to have complete records of course assessments and evaluations. If a student misses a quiz, midterm exam, or any other continuous assessment without a valid reason, they will receive a score of zero. However, if a student fails to attend the final exam or complete a term paper/project without a valid reason, the instructor will record an "NG" according to senate legislation, Art. 171.

It is important to carefully follow the instructions provided in your course guidebook for completing all requirement of the course.

12. GRADING SYSTEM

The gradingscaleofthecourseisfixedscaleaccordingtosenatelegislationofArt.177.

Table4: Grading scale of the course

Raw Mark Interval (100%)	Corresponding Letter Grade	Corresponding Fixed Number Grade
$90 \leq x \leq 100$	A ⁺	4.0
$85 \leq x < 90$	A	4.0
$80 \leq x < 85$	A ⁻	3.75
$75 \leq x < 80$	B ⁺	3.50
$70 \leq x < 75$	B	3.00
$65 \leq x < 70$	B ⁻	2.75
$60 \leq x < 65$	C ⁺	2.50
$50 \leq x < 60$	C	2.00
$45 \leq x < 50$	C ⁻	1.75
$40 \leq x < 45$	D	1.00
$x < 40$	F	0.00

13. TEXTBOOKS

- a. James M. Gere, Mechanics of Materials, 6th Edition

14. REFERENCES

- a. Beer, Johnston, De. Wolf, Mechanics of Materials, 4th Edition
- b. E.J. Hearn, Mechanics of Materials Volume I, 3rd Edition
- c. Egor P. Popov, Mechanics of Materials, 2nd Edition

15. APPROVAL

The following bodies confirmed the course guide book accordingly.

Position	Name	Signature	Date
Course Instructor	Dawit Molla		03/26/2025
Course Module Chair	Amare Mahtsentu		
Chair Holder (Mechanical Design)	Addisu Negash (PhD)		
V/Dean	Abebe Genetu		

Notes:

Student out comes and course learning out comes are related but distinct concepts in education:

1. **Student Outcomes:** These are the specific knowledge, skills, or attributes that students are expected to demonstrate or achieve by the end of a program or course. They focus on what students will be able to do or demonstrate as a result of their learning experiences. Student outcomes are often aligned with broader educational goals and may include abilities such as critical thinking, problem-solving, effective communication, and subject-specific skills.
2. **Course Learning Outcomes:** These are the specific statements that describe what students should be able to know or do as a result of completing a particular course. Course learning outcomes are more specific and focused on the content and objectives of the course itself. They outline the expected achievements or competencies that students should attain by the end of the course. These outcomes guide course design, instruction, assessment, and evaluation.

In summary, student out comes are broader and pertain to the overall goal so fan educational program, while course learning outcomes are more specific and relate directly to the content and objectives of individual courses within that program.

In education curriculum design, both student outcome performance indicators (PIs) and course learning out comes (CLOs) are important components, but they serve lightly different purposes:

1. **Student Outcome Performance Indicators (PIs):**
 - PIs a respecify, measurable criteria used to assess students' attainment of learning outcomes.
 - Theyprovidedetaileddescriptionsofwhatstudentsshouldbeabletodemonstrateor accomplish to meet the learning outcomes.
 - PIs often include observable behaviors, skills, or knowledge that can be assessed through various methods such as exams, assignments, projects, presentations, or observations.
 - Plshelpeducatorsandcurriculumdesignersdevelopassessmentstrategiesandtoolsto evaluate students' progress and achievement.
 - Pisareoftenalignedwithcourseobjectivesandstandardsensurethatassessment activities effectively measure student learning.
2. **Course Learning Outcomes (CLOs):**
 - CLOs are broadstatementsthatdescribewhatstudentsareexpectedtolearnor achieve by the end of a course or program.
 - Theyrepresenttheoverarchinggoalsandpurposesofthecourseandprovidea framework for instructional planning and assessment.
 - Close articulate the knowledge, skills, abilities, or attitudes that students should develop as a result of their learning experiences.
 - CLOs are typically derived from broad educational goals, program objectives, or professional standards.
 - CLOs guide the selection of instructional content, activities, and assessments to ensure that they are aligned with desired learning outcomes.

In summary, PIs are the specific criteria used to measure student achievement of CLOs. While CLOs provide a high-level overview of desired learning outcomes for a course or program, PIs offer more detailed, observable indicators of student performance that can be used to assess progress and attainment of those outcomes.